Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	13 October 2026 (documentation date)
Team ID	N/A — Individual project
Project Name	Employee Raise Issue Management System
Maximum Marks	4 Marks

Technical Architecture

The Employee Raise Issue Management System is implemented on the ServiceNow platform. Employees access a Service Portal page and use a Record Producer to submit an issue. The submitted information is mapped to a custom Employee Raise Issue table for tracking and processing.

USER / EMPLOYEE Opens the Employee Center / Service Portal and selects “Raise an Issue”.
PRESENTATION LAYER Service Portal page with Commercial Services and Raise an Issue custom widgets (HTML, CSS, client controller).
SUBMISSION & VALIDATION Record Producer: requester, category, subcategory, state and issue description. UI Policies enforce field behavior; dictionary dependency controls category/subcategory choices.
APPLICATION DATA Custom scoped application and Employee Raise Issue table; records are stored and managed in the ServiceNow platform database.
PLATFORM SERVICES ServiceNow-hosted platform provides authentication/access controls, record handling, attachments and platform availability.

Architecture notes: No separate external API, machine-learning model, or independently configured cloud database was implemented in this project. The application uses the data storage and hosting provided by ServiceNow.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employee-facing portal page and issue submission form.	ServiceNow Employee Center / Service Portal; HTML, CSS, JavaScript/AngularJS widget controller
2	Application Logic-1	Captures and submits employee issue details through the Record Producer.	ServiceNow Record Producer and catalog variables
3	Application Logic-2	Applies field requirements and behavior, including mandatory short description when State is New and read-only Number.	ServiceNow UI Policies and UI Policy Actions
4	Application Logic-3	Restricts subcategory options according to the selected category.	Dictionary dependent field and choice configuration
5	Database	Stores submitted issue records and their fields.	Custom ServiceNow table: Employee Raise Issue (x_2107432_employ_0_employee_raises_issue); ServiceNow platform database
6	Cloud Database	Database service is managed within the ServiceNow platform; no separate cloud database was configured.	ServiceNow-managed platform storage
7	File Storage	Allows users to attach supporting files to a submitted issue, where enabled.	ServiceNow attachment functionality
8	External API-1	No external API was implemented.	Not applicable
9	External API-2	No second external API was implemented.	Not applicable
10	Machine Learning Model	No machine-learning model is part of the current implementation.	Not applicable
11	Infrastructure (Server / Cloud)	The application is developed and hosted on a ServiceNow developer instance and uses the platform's hosting services.	ServiceNow cloud platform; instance URL used for development/testing

Table-2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	The widget uses HTML, CSS and client-side JavaScript in the Service Portal. No separate open-source application framework was added as part of this project.	HTML, CSS, JavaScript; ServiceNow Service Portal widget framework
2	Security Implementations	Access is handled through the ServiceNow platform's authentication and role-based access model. No custom encryption algorithm or firewall configuration was implemented by the project.	ServiceNow platform authentication, roles and access controls
3	Scalable Architecture	The solution uses a platform-hosted application and a custom table, allowing records to be managed within ServiceNow. No separate microservices architecture was built.	ServiceNow platform architecture
4	Availability	The application relies on the availability of the ServiceNow	ServiceNow-hosted instance

		instance and its platform services. No custom load balancer or distributed server setup was configured.	
5	Performance	The form uses catalog variables, UI Policies and table-level choice configuration to standardize submission. No custom caching, CDN, or throughput benchmark was implemented.	ServiceNow platform and native form/widget behavior

References

- ServiceNow developer instance used to build and test the project.
- ServiceNow Service Portal and Record Producer configuration used in the implementation.
- Project implementation: Employee Raise Issue custom table, UI Policies, dependent choices, Record Producer and custom portal widgets.

Note: This document describes the features implemented in the project. Confirm the date and exact table technical name against your instance before final submission.